

The 2×2 upper-bild trichotomy fails in every dimension $n \geq 3$

Candidate solution packet for arXiv:2008.02850

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Status. Candidate full negative answer, likely valid, awaiting specialist review. The source paper already treats two regimes at size three; the new ingredient is the missing singular positive-semidefinite regime, including its exact upper bild, followed by a uniform closure of all three regimes for every $n \geq 3$.

1 Source question and answer

Carvalho, Diogo, and Mendes ask after Corollary 3.11 of [1] (source PDF, p. 9) whether any part of its 2×2 description can be generalized to $n > 2$:

Theorem [3.6](#) generalizes theorem 6.1 of the outstanding *tour de force* [ST](#). In the next proposition it is established that for 2×2 complex matrices the upper bild is either $B^+(A) = \text{conv}\{W_{\mathbb{C}}^+(A), W_{\mathbb{C}}^+(A^*)\}$, $B^+(A) = \text{conv}\{W_{\mathbb{C}}^+(A), v\}$, for some $v \in \mathbb{R}$, or $B^+(A) = W_{\mathbb{C}}^+(A)$. We thus provide a different and more straightforward proof for the above mentioned theorem of [ST](#) (see the appendix for the proof).

Corollary 3.11. *Let $A = H + Si \in \mathcal{M}_2(\mathbb{C})$. The upper bild of A is:*

- (i) $B^+(A) = \text{conv}\{W_{\mathbb{C}}^+(A), W_{\mathbb{C}}^+(A^*)\}$, if S is indefinite;
- (ii) $B^+(A) = \text{conv}\{W_{\mathbb{C}}^+(A), v\}$, for some real value v , if S is positive definite;
- (iii) $B^+(A) = W_{\mathbb{C}}^+(A)$, if S is positive semi-definite.

A natural question to ask is if the previous corollary, or part of it, can be generalized for $n > 2$. That is, when the matrix S is positive definite, positive semi-definite or indefinite, can the shape of the quaternionic numerical range given by theorem [3.6](#) be further simplified? Next example shows that it is not the case for 3×3 normal complex matrices.

Example 3.12. Consider the complex matrix $A = H + Si \in \mathcal{M}_3(\mathbb{C})$, where $A = \text{diag}(1+i, 1+i, -i)$. Hence $W_{\mathbb{C}}(A) = [1+i, -i]$ and $\text{conv}\{W_{\mathbb{C}}^+(A), W_{\mathbb{C}}^+(A^*)\} = \text{conv}\{i, 1/2, 1+i\}$. Since matrix A is normal, its upper bild is $\text{conv}\{i, 1/2, 1, 1+i\}$. We have, then, an example of a matrix A , with an indefinite matrix S , $S = \text{diag}(1, 1, -1)$, where $B^+(A) \neq \text{conv}\{W_{\mathbb{C}}^+(A), W_{\mathbb{C}}^+(A^*)\}$.

More precisely, for $A = H + Si$ they ask whether the upper bild can still be simplified according as S is positive definite, positive semidefinite, or indefinite. The three formulas displayed in Corollary 3.11 are

$$\begin{array}{ll} B^+(A) = \text{conv}\{W_{\mathbb{C}}^+(A), W_{\mathbb{C}}^+(A^*)\} & (S \text{ indefinite}), \\ B^+(A) = \text{conv}\{W_{\mathbb{C}}^+(A), v\} \quad (v \in \mathbb{R}) & (S \text{ positive definite}), \\ B^+(A) = W_{\mathbb{C}}^+(A) & (S \text{ positive semidefinite}). \end{array}$$

Because the second and third hypotheses overlap in standard terminology, we read the last branch in the intended trichotomy as singular positive semidefinite. The answer is negative in every regime and every dimension $n \geq 3$.

2 Definitions

For $A \in M_n(\mathbb{C}) \subset M_n(\mathbb{H})$, let

$$W_{\mathbb{H}}(A) = \{x^*Ax : x \in \mathbb{H}^n, x^*x = 1\}.$$

Writing $\mathbb{C}^+ = \{a + bi : a, b \in \mathbb{R}, b \geq 0\}$, its upper bild is $B^+(A) = W_{\mathbb{H}}(A) \cap \mathbb{C}^+$. We similarly write $W_{\mathbb{C}}^+(A) = W_{\mathbb{C}}(A) \cap \mathbb{C}^+$. Quaternionic units satisfy $i^2 = j^2 = k^2 = ijk = -1$; in particular

$$j^*ij = -i, \quad j^*(1+i)j = 1-i. \quad (1)$$

3 Proof intuition

The source's two examples exploit two copies of the eigenvalue $1+i$. Quaternionic conjugation by j reverses its imaginary part, so averaging the two copies produces the real point 1. This point lies outside the complex hulls proposed in the indefinite and positive-definite branches.

To expose the missing singular-PSD branch, retain those two copies and add a zero eigenvalue. The same average produces 1, already outside the complex segment $[0, 1+i]$. More is true. If the first two squared coordinate norms sum to a , their imaginary contributions have total norm at most a . Conversely, splitting those weights as $(a+b)/2$ and $(a-b)/2$ in quaternionic directions 1 and j produces $a+bi$. This identifies the entire upper bild as a triangle. Repeating the final diagonal entry then gives examples in every higher dimension without changing the obstruction.

4 The uniform counterexamples

Theorem 1. *For every integer $n \geq 3$, set*

$$\begin{aligned} A_n^0 &= \text{diag}(1+i, 1+i, 0, \dots, 0), \\ A_n^+ &= \text{diag}(1+i, 1+i, i, \dots, i), \\ A_n^\pm &= \text{diag}(1+i, 1+i, -i, \dots, -i). \end{aligned}$$

Then:

1. $A_n^0 = H + Si$ has singular positive-semidefinite S , and

$$W_{\mathbb{C}}^+(A_n^0) = [0, 1+i], \quad B^+(A_n^0) = \text{conv}\{0, 1, 1+i\}.$$

In particular $B^+(A_n^0) \neq W_{\mathbb{C}}^+(A_n^0)$.

2. $A_n^+ = H + Si$ has $S = I_n > 0$, but there is no $v \in \mathbb{R}$ such that

$$B^+(A_n^+) = \text{conv}\{W_{\mathbb{C}}^+(A_n^+), v\}.$$

3. $A_n^\pm = H + Si$ has indefinite S , but

$$B^+(A_n^\pm) \neq \text{conv}\{W_{\mathbb{C}}^+(A_n^\pm), W_{\mathbb{C}}^+((A_n^\pm)^*)\}.$$

Consequently none of the three formulas in Corollary 3.11 extends universally to any dimension $n \geq 3$.

The first assertion follows from an exact calculation.

Lemma 2. For $n \geq 3$,

$$B^+(A_n^0) = \{a + bi : 0 \leq b \leq a \leq 1\} = \text{conv}\{0, 1, 1 + i\}.$$

Proof. Let $q = (q_1, \dots, q_n) \in \mathbb{H}^n$ be a unit vector. Write $q_\ell = r_\ell u_\ell$ for $\ell = 1, 2$, where $r_\ell \geq 0$ and u_ℓ is a unit quaternion (with an arbitrary choice if $r_\ell = 0$). Only the first two diagonal entries contribute, so

$$q^* A_n^0 q = a + r_1^2 u_1^* i u_1 + r_2^2 u_2^* i u_2, \quad a = r_1^2 + r_2^2 \in [0, 1].$$

Each $u_\ell^* i u_\ell$ is a unit pure imaginary quaternion. Hence the norm of the imaginary part is at most $r_1^2 + r_2^2 = a$. If the displayed value lies in $\mathbb{C}^+$, it has the form $a + bi$ with $0 \leq b \leq a$.

Conversely, take $0 \leq b \leq a \leq 1$ and define

$$q = \left(\sqrt{\frac{a+b}{2}}, \sqrt{\frac{a-b}{2}} j, \sqrt{1-a}, 0, \dots, 0 \right).$$

This is a unit vector, and (1) gives

$$\begin{aligned} q^* A_n^0 q &= \frac{a+b}{2}(1+i) + \frac{a-b}{2}(1-i) \\ &= a + bi. \end{aligned}$$

Thus every point in the triangle is attained. □

Proof of Theorem 1. The matrix A_n^0 is normal, so its complex numerical range is the convex hull of its eigenvalues, namely $[0, 1 + i]$. Lemma 2 proves item 1, including strictness because 1 belongs to the triangle but not to the diagonal segment.

For A_n^+ , the imaginary-part matrix is I_n , and normality gives

$$W_{\mathbb{C}}^+(A_n^+) = W_{\mathbb{C}}(A_n^+) = [i, 1 + i].$$

Using only the first three coordinates and (1), the unit vectors

$$q = 2^{-1/2}(1, j, 0, \dots, 0), \quad p = 2^{-1/2}(1, 0, j, 0, \dots, 0)$$

give

$$q^* A_n^+ q = 1, \quad p^* A_n^+ p = \frac{1}{2}.$$

Thus $B^+(A_n^+) \cap \mathbb{R}$ contains two distinct points. On the other hand, for any real v , every point of $\text{conv}\{[i, 1 + i], v\}$ is $tz + (1 - t)v$ with $z \in [i, 1 + i]$ and $0 \leq t \leq 1$; its imaginary part is t . Therefore that hull meets $\mathbb{R}$ only at v . Item 2 follows.

Finally, the imaginary-part matrix of $A_n^\pm$ has both signs. Its complex numerical range and that of its adjoint are respectively

$$[-i, 1 + i] \quad \text{and} \quad [i, 1 - i].$$

Taking upper halves yields

$$\text{conv}\{W_{\mathbb{C}}^+(A_n^\pm), W_{\mathbb{C}}^+((A_n^\pm)^*)\} = \text{conv}\{i, \frac{1}{2}, 1 + i\}.$$

This triangle meets the real axis only at $1/2$. Yet the same vector $q = 2^{-1/2}(1, j, 0, \dots, 0)$ gives $q^* A_n^\pm q = 1$. Hence 1 lies in the actual upper bild and outside the proposed hull, proving item 3. □

5 Verification, scope, and novelty

The exact quaternion identities and the symbolic parametrization in Lemma 2 are checked in `code/verify_counterexamples.py`. That script also performs an independent random-vector check of the norm bound. The latter is a sanity check only; the proof is the triangle inequality and explicit construction above.

The claim answers the source question in its intended concrete sense: none of the three formulas in the 2×2 trichotomy is a universal formula in any higher dimension. It does not rule out other, differently formulated simplifications. Example 3.12 of the source already supplies the $n = 3$ indefinite and positive-definite failures. The new mathematical content is the singular-PSD failure, its exact upper bild, and the dimension-uniform three-family closure.

A bounded novelty search through 13 August 2026 used the source title and arXiv id, exact matrix and formula searches, general web search, the run registry, and local title/abstract/full-source indexes. It found no separate occurrence of the singular-PSD triangle or the full uniform statement. This is not an exhaustive priority search.

6 Human-review recommendation

The result is elementary and likely valid. Review should begin with Lemma 2, especially the assertion that conjugation of i by a unit quaternion preserves the norm of the pure imaginary part and the explicit converse vector. Novelty should be judged as a focused completion: two branches were already known from the source paper.

References

- [1] L. Carvalho, C. Diogo, and S. Mendes, *Quaternionic numerical range of complex matrices*, Linear Algebra Appl. **620** (2021), 168–181; arXiv:2008.02850.